

Exploratory Data Analysis: Customer Churn Project Report

Dataset Scope: 7,043 Unique Accounts | 21 Attributes (3 Numeric, 17 Categorical, 1 Unique Identifier)

1. Introduction & Executive Summary

I have just completed an Exploratory Data Analysis (EDA) on a customer churn dataset consisting of 7,043 unique rows and 21 architectural columns to identify what drives customer attrition. Through data preprocessing, cleaning, and visualizations, I have uncovered key trends that directly impact user retention.

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From my analysis, the overall portfolio stands at **5,174 retained customers (73.46%)** and **1,869 churned customers (26.54%)**.

2. Data Cleaning & Preprocessing Workflow

Before running the statistical and visual analytics, I performed a multi-step data engineering process to ensure the dataset was pristine and mathematically sound:

- **Structural Review:** I checked the initial shapes and dimensions, verifying an active matrix of 7,043 records across 21 discrete columns.
- **Handling Blank Values:** I identified that the TotalCharges column was classified as a generic string/object type because it contained blank spaces (" "). I successfully cleaned this column by replacing the empty spaces with a baseline value of 0 and converting the data type into a float (float64) using `astype("float")`.
- **Null & Duplicate Verification:** I ran programmatic checks via `.isnull().sum().sum()` and `.duplicated().sum()`. This verified **0 null entries** and **0 duplicate records** across the rows. I also verified that there were **0 duplicates** for the unique customerID attribute, confirming each account is distinct.
- **Feature Transformation:** The SeniorCitizen feature was originally stored as an integer flag (0 or 1). To make it easier to read and consistent with other categorical features, I built a custom mapping function and ran it across the column via `.apply()` to convert the numeric flags into descriptive No and Yes categories.

3. Financial Benchmarks & Lifespan Statistics

Running descriptive statistics (.describe()) over the numeric fields allowed me to map out the financial realities of the subscriber base:

- **Monthly Charges Portfolio:** The client network generates an average monthly revenue of **\$64.76** per account, with a median midpoint sitting at **\$70.35**. The top 25% premium billing bracket ranges from **\$89.85 up to \$118.75** every single month.
- **Accumulated Lifecycle Billing:** The historical lifetime financial throughput results in an average total charge of **\$2,279.73** per customer, with the most valuable account accumulating **\$8,684.80**.
- **Tenure Volatility:** The absolute baseline lifespan average across the data is **32.37 months**. However, my analysis shows severe instability at the beginning of the lifecycle: the bottom 25% of customers churn or experience volatility within their first **9 months** of service.

4. Cohort & Service Portfolio Deep Dive

By building target-focused visual count plots and aggregate grouping matrices, I systematically evaluated how different customer attributes and product selections relate to churn:

Demographic Cohort Vulnerabilities

- **Gender Separation:** Gender is highly balanced, containing **3,555 Male** and **3,488 Female** instances. When plotting churn against gender, I observed that **939 Females** and **930 Males** left the company. This nominal 9-account delta proves that **gender exerts zero statistical influence on churn**.
- **Senior Citizen Realities:** While seniors represent just **16.21%** (1,142 instances) of the subscriber footprint, my count plot shows a high churn density among them. Seniors leave at a rate significantly higher than younger cohorts, identifying them as a high-risk group.

Contractual & Billing Friction Points

- **Contract Length (The Leading Indicator):** My contract count plot shows that **Month-to-month contracts suffer from extreme churn**. They account for **1,655 total churned cases**, which represents **88.55% of all churned customers** in the dataset. In contrast, One-year contracts (166 churns) and Two-year contracts (only 48 churns) offer reliable business stability.
- **Payment Architecture:** Users paying via **Electronic Check** show a massive churn spike, contributing **1,071 instances** (making up **57.30% of total portfolio losses**). Conversely, automatic credit card (232 churns) and bank transfer payment paths (258 churns) maintain highly stable retention levels.

Product Portfolio Influence

Through my service analysis plots, I categorized features based on their relationship with customer churn:

Service Feature	Low-Churn Retention Drivers	High-Churn Vulnerabilities	Direct Analytical Insights
Internet Service	DSL	Fiber Optic	Fiber Optic lines show the highest churn volume despite offering faster speeds, suggesting potential pricing or connection issues.
Online Security	"Yes" Subscription	"No" Service	IPYNB Lacking online protection correlates with a dramatic increase in churn.
Online Backup	"Yes" Subscription	"No" Service	IPYNB Customers without decentralized cloud storage leave at much higher rates.
Device Protection	"Yes" Subscription	"No" Service	IPYNB Accounts without active hardware protection plans drop off quickly.
Tech Support	"Yes" Subscription	"No" Service	IPYNB Customers lacking access to technical help show high attrition rates.
Streaming TV/Movies	Balanced Distribution	Balanced Distribution	IPYNB Entertainment add-ons show an even distribution, meaning they don't significantly drive or prevent churn.
Phone/Multiple Lines	Balanced Distribution	Balanced Distribution	IPYNB Core phone configurations have a neutral effect on customer attrition.
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5. Strategic Data-Driven Recommendations

Based on the programmatic trends I uncovered in this analysis, I recommend implementing the following targeted operations to mitigate customer churn:

1. Address Month-to-Month Contract Vulnerabilities

- **Insight:** Over 88% of all lost accounts belong to Month-to-month contracts.
- **Action:** Create an automated "Migration Promotion" that flags Month-to-month accounts reaching their 6th or 9th monthly billing cycle (the critical early drop-off window). Offer a small monthly discount or service upgrade (such as free Streaming features) conditional on moving to a stable 1-Year contract.

2. Transition Electronic Check Users to Automated Billing

- **Insight:** Electronic Check transactions are highly correlated with churn, accounting for 57.30% of portfolio losses.
- **Action:** Phase out this friction point by introducing a small "manual handling fee" for check payments. Simultaneously, offer a one-time \$5 bill credit to incentivize customers to switch to automatic payment methods via credit card or bank transfer, which have much higher retention rates.

3. Package Tech Support and Security as Standard Features

- **Insight:** Customers without specialized value-added features like Tech Support, Online Security, and Device Protection exhibit the highest churn rates[cite: 1].
- **Action:** Instead of selling these security options as separate add-ons, bundle them directly into core internet plans, particularly for high-churn Fiber Optic accounts[cite: 1]. Marketing these services as built-in benefits increases product utility and reduces early account attrition